Pneumonia Detection from Chest X-Rays

Deep-Learning Screening of Pediatric Chest Radiographs — NORMAL vs PNEUMONIA

Project Report — Business Framing, Data Quality, EDA, Modeling, Training History & Results

99.0%	98.6%	2
Best val recall — ResNet50	Val accuracy — ResNet50	deep-learning models trained

Primary KPI: **Recall** (missing pneumonia is the dangerous error) — Recall > F1 > AUC > Accuracy.

1. Business Problem

Hospitals receive large volumes of chest X-rays daily and radiologists are overloaded, which can delay pneumonia diagnosis. A **missed** pneumonia case is the costly error, so this project builds an automated screen that triages radiographs as NORMAL vs PNEUMONIA, flags high-risk patients first, and reduces missed cases.

Business value

- **Faster diagnosis** — seconds per image vs minutes of manual screening.
- **Triage** — auto-flag likely-pneumonia X-rays for urgent review.
- **Reduced workload** — radiologists focus on ambiguous cases.
- **Consistent screening** — uniform criteria, fewer missed cases.

Success criteria

Because the hospital objective is to **minimise false negatives**, the primary KPI is **Recall** on the PNEUMONIA class, then F1 > AUC > Accuracy. Models are checkpointed on the best validation recall, the 2.69:1 class imbalance is countered with class-weighted loss, and a recall-first decision threshold is preferred over a naive 0.5.

2. Dataset & Data Quality

Splits are stored as non-destructive CSV manifests (5,830 images total). The original validation set had only 16 images, so train+val were merged and stratified-resplit (15% validation); the official TEST set is left untouched for comparability.

Split	NORMAL	PNEUMONIA	Total
train	1,146	3,279	4,425
val	202	579	781
test	234	390	624
all	**1,582**	**4,248**	**5,830**

Data cleaning

- **26 exact-duplicate images** detected (content hash) and excluded from the train/val pool.
- **0 corrupt/empty files**.
- Images are JPEG at variable resolution, standardised to 224×224 for modeling.

3. Exploratory Data Analysis

3.1 Class distribution & balance

About **72.9%** of images are PNEUMONIA (a 2.69:1 imbalance), which motivates class-weighted loss and augmentation and makes recall — not accuracy — the right headline metric.

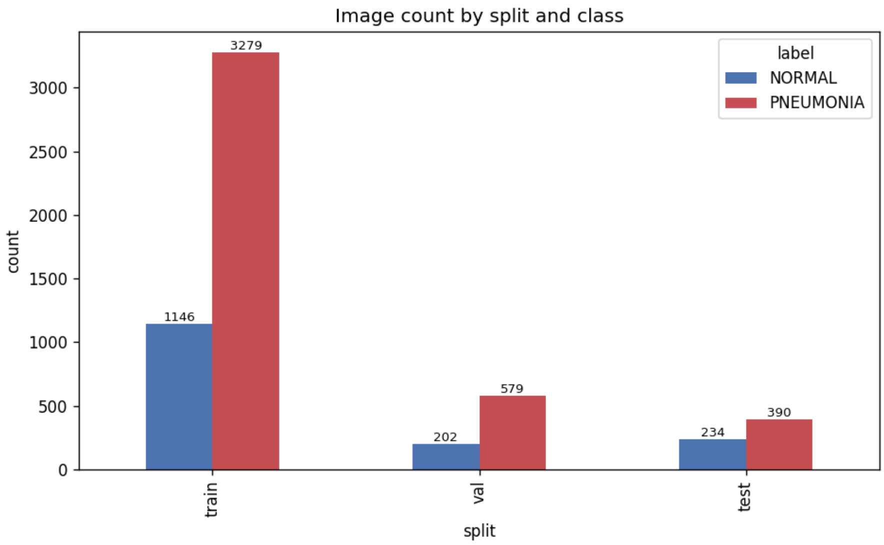

Fig 1. Image counts per class.

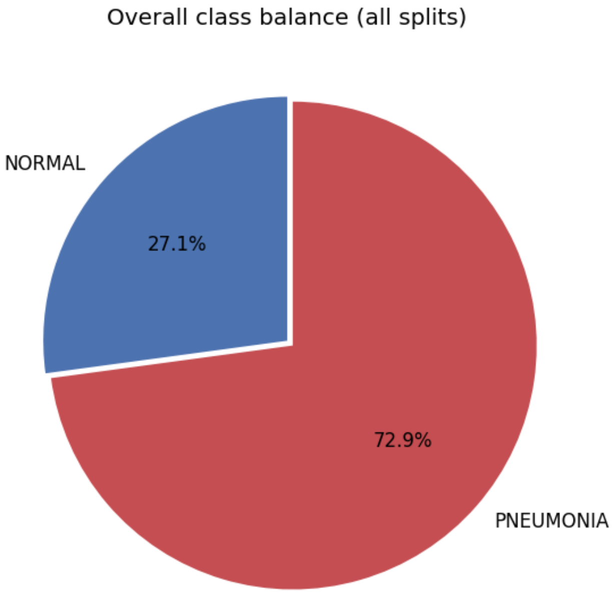

Fig 2. Overall class balance (PNEUMONIA 72.9%).

3.2 Sample radiographs

NORMAL X-rays show clear lung fields; PNEUMONIA X-rays show white opacities and dense infiltrates.

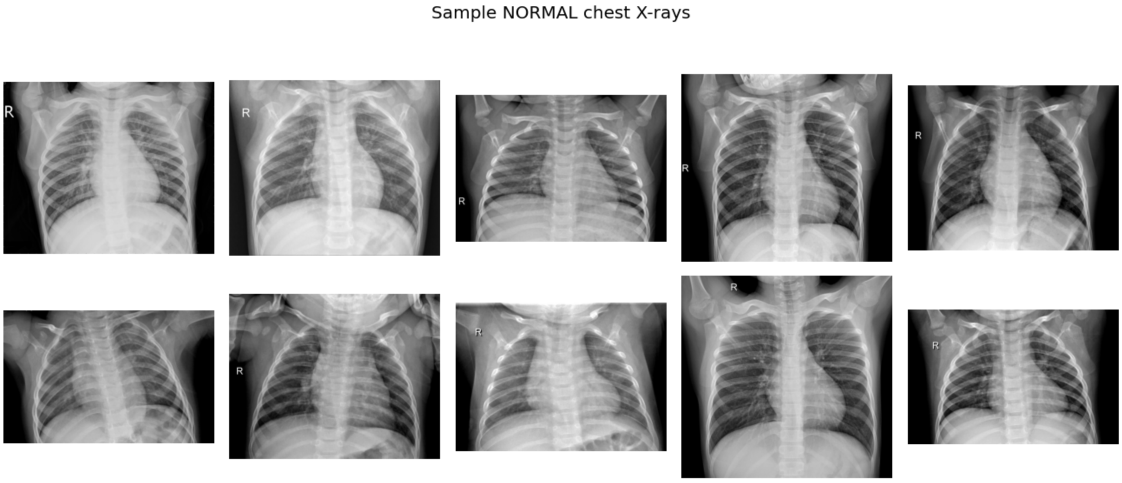

Fig 3. Sample NORMAL chest X-rays (clear lungs).

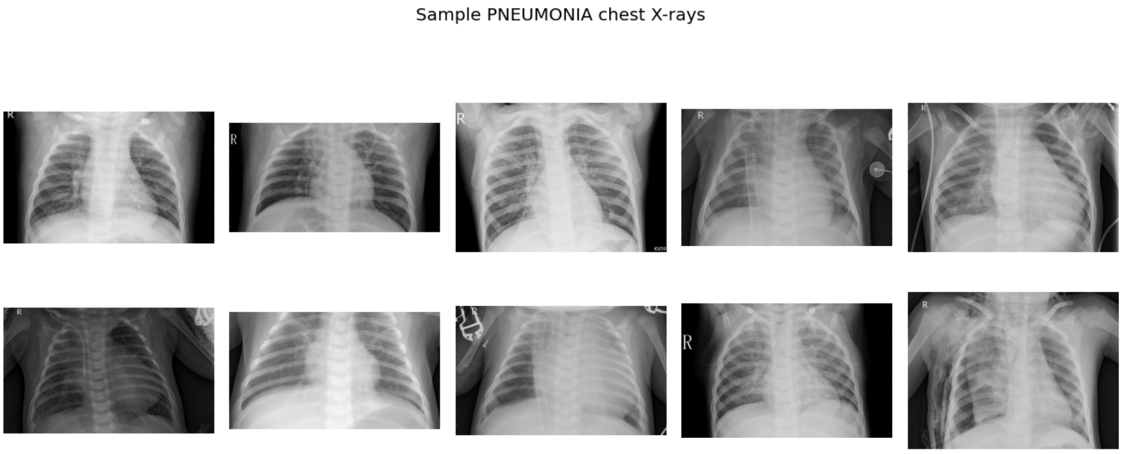

Fig 4. Sample PNEUMONIA X-rays (opacities / infiltrates).

3.3 Image size & pixel intensity

Resolutions vary widely (width 384–2916 px, height 127–2713 px, median ~1280×886), so all images are resized to 224×224. Pneumonia images skew brighter, reflecting denser opacities.

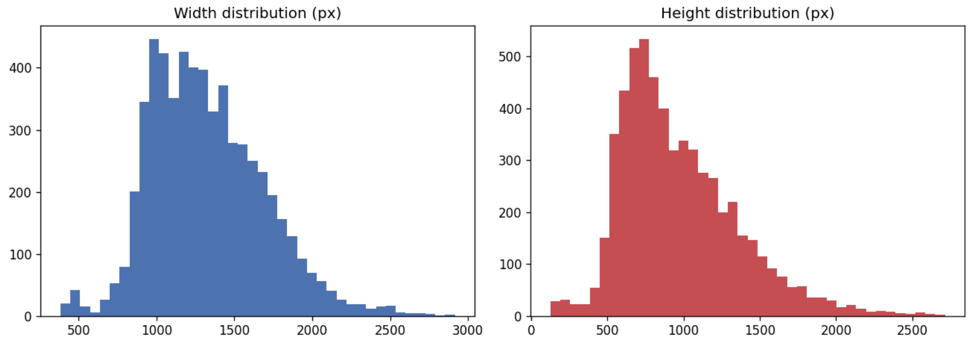

Fig 5. Distribution of image dimensions.

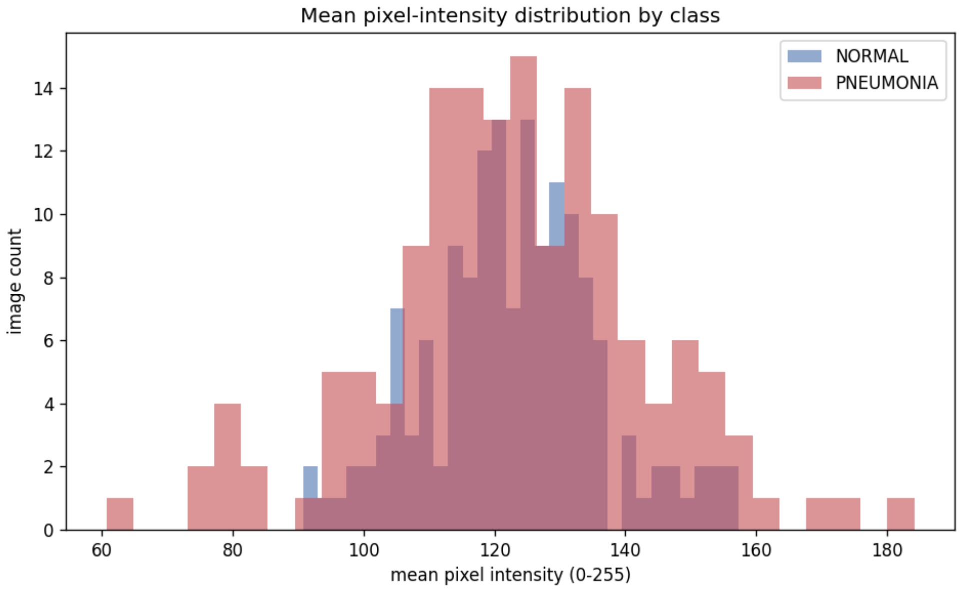

Fig 6. Mean pixel-intensity distribution by class.

4. Modeling Approach

Two architectures were trained and compared, both checkpointed on best validation recall with class-weighted cross-entropy to counter the imbalance:

Model 1 — Custom CNN (baseline)

A compact convolutional baseline (Conv-BN-ReLU-Pool blocks $\rightarrow$ global average pooling $\rightarrow$ classifier) trained from scratch to establish a reference point.

Model 2 — ResNet50 (transfer learning)

An ImageNet-pretrained ResNet50 backbone with a new 2-class head, fine-tuned in two phases: freeze the backbone and train the head first, then unfreeze and fine-tune the whole network at a lower learning rate. Grayscale X-rays are replicated to 3 channels to reuse the pretrained weights.

Training setup

- Class-weighted cross-entropy (inverse frequency) for the 2.69:1 imbalance.
- Checkpoint the model with the best **validation recall**.
- Augmentation (rotation, zoom, flip, brightness) on the training split only.
- ImageNet normalisation; images standardised to 224×224.

5. Training History

Validation recall, precision, F1, AUC and loss were tracked each epoch; the checkpoint is the best-recall epoch.

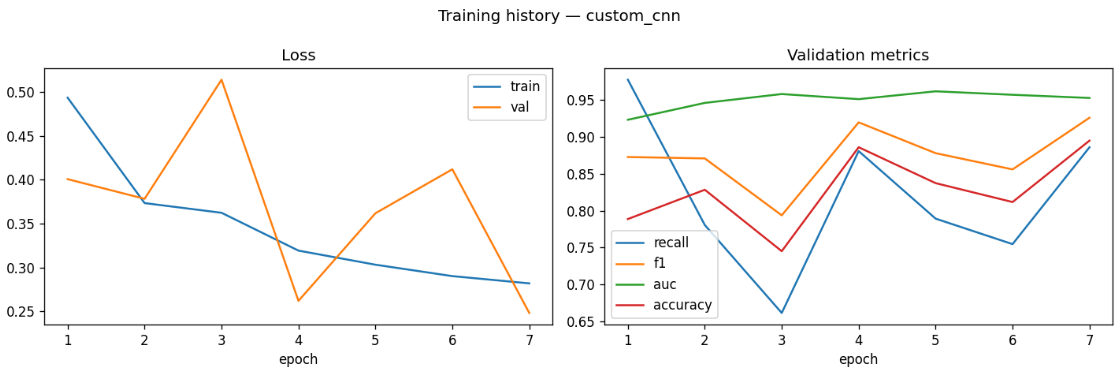

Fig 7. Custom CNN training history.

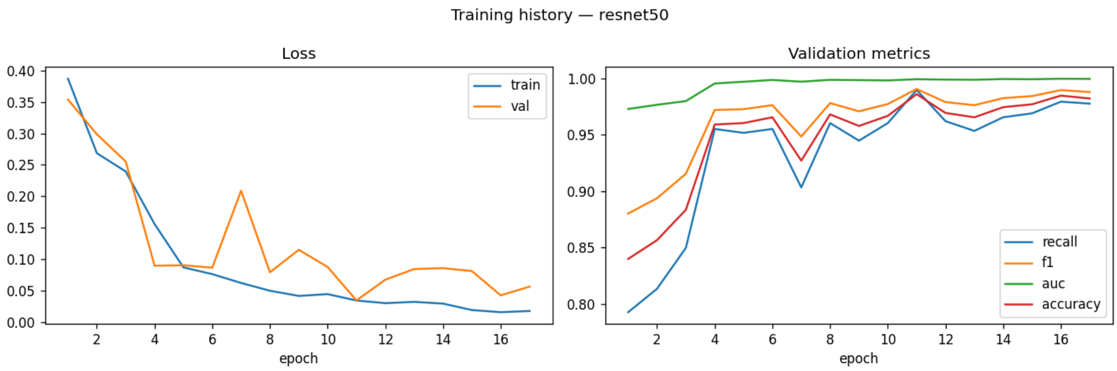

Fig 8. ResNet50 training history.

6. Results & Model Comparison

Metrics below are from each model's **best validation-recall epoch** (the checkpoint-selection metric). The best-recall model leads the table.

Model	Recall	Precision	F1	AUC	Accuracy	Best epoch	Train min
ResNet50	**0.990**	**0.991**	**0.990**	**0.999**	**0.986**	**11/17**	**6.71**
Custom CNN	0.978	0.788	0.873	0.923	0.789	1/7	3.42

Best model: ResNet50 — validation recall 0.990, F1 0.990, AUC 0.999, accuracy 0.986 at epoch 11/17. Transfer learning with ResNet50 outperforms the from-scratch baseline on the recall-first objective.

7. Business Impact & Future Work

Business impact

- **High recall** — the best model reaches 99.0% validation recall, keeping missed-pneumonia cases low.
- **Triage** — auto-flag likely-pneumonia X-rays for urgent review.
- **Speed & consistency** — seconds per image, uniform screening.
- **Recall-first threshold** keeps the false-negative rate low at deployment.

Future work

- Add a third backbone (e.g. EfficientNet) and an ensemble.
- Evaluate on the held-out TEST set with a recall-tuned threshold and report confusion / ROC / PR curves.
- Grad-CAM explainability to show which lung regions drive predictions.
- Calibrated probabilities with abstention for borderline cases.
- Deployment behind an API with PACS/RIS integration.

Generated from real project artifacts in `outputs/`. Per-model metrics are the best-validation-recall epoch of each history CSV; figures are embedded only where they exist on disk.